

Second Addendum for *MISO DPP 2016 February West Area* *Phase 3 Study Final*

February 20, 2020

This addendum contains updated study results for the report entitled MISO DPP 2016 February West Area Phase 3 Study and dated March 5, 2018. Results have been updated to reflect Network Upgrade (NU) changes after the withdrawal of generation project J528 (200 MW). The withdrawal of this project does not trigger a restudy for remaining DPP-2016-February West area projects.

1.1 Withdrawal of J528

Withdrawal of generation project J528 was studied to determine continued need for network upgrades and calculate update cost allocation percentages. After the withdrawal of J528, the following Network Upgrades (NUs) are determined to still be needed:

Table 1: Thermal Network Upgrades Needed in MISO Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Bondurant-Montezuma 345 kV	926.0	MEC	921.5	91	CEII	P0	J432, J460, J488, J493, J495, J498, J499, J500, J506, J514, J524, J526, J526, J527, J534, J535
Bondurant-Montezuma 345 kV	1010.0	MEC	995.2	99	CEII	P1	J499, J500, J527
Bondurant-Montezuma 345 kV	1010.0	MEC	1029.3	103	CEII	P2-P7	J499, J500, J527

Table 2: Voltage Network Upgrades Needed in MISO Analysis

Mitigation	Contingency	Monitored Element	
Montezuma 100 Mvar Capacitor Bank		Bus	J528 Withdrawn VCONT
	CEII	635730 Montezuma	1.0005
	CEII	75730 J530 POI	1.0029
J438 25 Mvar Capacitor Bank		Bus	J528 Withdrawn VCONT
	CEII	51113 J438 POI	1.0006
Blackhawk 150 Mvar Capacitor Bank		Bus	J528 Withdrawn VCONT
	CEII	636199 Blackhawk	0.9377
	CEII	636199 Blackhawk	0.941
Midport 50 Mvar Capacitor Bank		Bus	J528 Withdrawn VCONT
	CEII	636240 Butler	0.948

Network Upgrade	Cost	NUs Type
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	Reactive Power NU

The upgrades that are still required even with the withdrawal of J528 are to be reallocated to the remaining projects by the percentages documented in the table below.

Table 3: Network Upgrades ERIIS Cost Allocation

Mitigation	FAS NU Cost	J432	J460	J475	J485	J488	J493	J495	J498
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,045,000	\$90,670	\$176,742	\$0	\$0	\$133,333	\$137,080	\$162,958	\$753,842
Bondurant-Montezuma Structure Remediation	\$320,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Mitigation	FAS NU Cost	J499	J500	J504	J506	J510	J514	J523	J524
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,045,000	\$959,178	\$1,329,924	\$0	\$264,925	\$0	\$26,327	\$0	\$75,466
Bondurant-Montezuma Structure Remediation	\$320,000	\$103,524	\$143,472	\$0	\$0	\$0	\$0	\$0	\$0
Mitigation	FAS NU Cost	J526	J527	J528	J529	J530	J534	J535	
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,045,000	\$272,889	\$677,019	\$0	\$297,134	\$0	\$324,521	\$362,992	
Bondurant-Montezuma Structure Remediation	\$320,000	\$0	\$73,004	\$0	\$0	\$0	\$0	\$0	

The withdrawal of J528 did result in the determination that certain Network Upgrades were no longer required for the 2016 February West cycle.

Table 4: ERIIS Upgrades Determined to be no longer needed

Network Upgrade	Cost	NUs Type
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	Reactive Power NU

1.2 Conclusion

A restudy is determined to be not needed for the remaining DPP-2016-February West area projects. The updated cost allocation for the ERIS upgrades that are still required were calculated. Network Upgrades that are no longer required were also determined.